THE TAME SYMBOL IS WELL DEFINED INDEPENDENTLY OF THE FACTORIZATIONS AND OF THE CHOSEN EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed. The author's notation (1)-(6) list inside the proof and the phrase 'This proves the desired equality up to signs' are reproduced verbatim.

Editorial note. source_id and source_version carry the current authoritative snapshot label of this source, 20261012, read from sources/stacks-chow/license-observation.json and SOURCE.json at the moment of opening.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros) of sources/stacks-chow/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. Three of the referenced results are same-chapter and are located at their \label lines inside the pinned chapter file; the two cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt).

Allowed standard prerequisites (editorial).

- The displayed formula defining the tame symbol, and the preparation lemma producing local factorizations. Same chapter, the displayed formula labelled equation-tame-symbol (source line 820, Stacks tag 0EAQ) and Lemma labelled lemma-prepare-tame-symbol (source line 642, Stacks tag 0EAG); named in the counted statement at line 831 and invoked at proof line 865. Supply the formula whose well-definedness is asserted and the class of extensions B over which the reader is allowed to compute; the two independence arguments are the counted content.
- Transitivity of the norm in a tower of field extensions. Fields, Lemma labelled fields-lemma-trace-and-norm-tower (canonical Stacks tag 0BIJ); invoked at proof lines 875-876. Supplies the reduction of the independence of B to the numerical identity lambda^m = prod Norm(lambda)^{m_k}.
- Length bookkeeping for a finite extension of one-dimensional local rings. Same chapter, Lemma lemma-compare-periodic-lengths (source line 283, Stacks tag 0EA9) with the equation equation-multiplicity-coker-ker (source line 203, Stacks tag 0EA6), and Algebra, Lemma algebra-lemma-pushdown-module (canonical Stacks tag 02M0); invoked at proof lines 894-898. Supply m = length_B(C/pi C) = sum m_k [kappa_k : kappa], the equality that makes the displayed norm identity true.

Source reference. Original equation 0EAQ: sources/stacks-chow/pinned-original-source.tex, source lines 820–820. Same-chapter pointer to the displayed definition

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of the tame symbol, named in the counted statement at line 831; both independence proofs are retained in the counted span.

Source reference. Original lemma 0EAG: sources/stacks-chow/pinned-original-source.tex, source lines 642–642. Same-chapter pointer invoked at proof line 865 for the class of admissible extensions; the reduction to the norm identity is retained in the counted span.

Source reference. Original lemma 0BIJ: sources/stacks-chow/tags.txt, source lines 14492–14492. Cross-chapter standard prerequisite invoked once at proof lines 875-876 for transitivity of norms.

Source reference. Original lemma 0EA9: sources/stacks-chow/pinned-original-source.tex, source lines 283–283. Same-chapter pointer invoked at proof line 895 for the length comparison m = length_B(C/pi C); the final assembly of the identity is retained in the counted span.

Source reference. Original equation 0EA6: sources/stacks-chow/pinned-original-source.tex, source lines 203–203. Same-chapter pointer to the defining display of e_R, invoked together with the comparison lemma at proof line 896.

Source reference. Original lemma 02M0: sources/stacks-chow/tags.txt, source lines 3427–3427. Cross-chapter standard prerequisite invoked once at proof line 898 for length_B(C/pi C) = sum m_k [kappa_k : kappa], the last numerical input.

Lemma 0.1. *The formula (0EAQ) determines a well defined element of $\kappa(\mathfrak{m})^*$. In other words, the right hand side does not depend on the choice of the local factorizations or the choice of B.*

Proof. Independence of choice of factorizations. Suppose we have a Noetherian 1-dimensional local ring B, elements $a_1, a_2 \in B$, and nonzerodivisors π, θ such that we can write

$$a_1 = u_1\pi^{e_1} = v_1\theta^{f_1}, \quad a_2 = u_2\pi^{e_2} = v_2\theta^{f_2}$$

with $e_i, f_i \geq 0$ integers and u_i, v_i units in B. Observe that this implies

$$a_1^{e_2} = u_1^{e_2}u_2^{-e_1}a_2^{e_1}, \quad a_1^{f_2} = v_1^{f_2}v_2^{-f_1}a_2^{f_1}$$

On the other hand, setting $m = \text{length}_B(B/\pi B)$ and $k = \text{length}_B(B/\theta B)$ we find $e_2 m = \text{length}_B(B/a_2B) = f_2 k$. Expanding $a_1^{e_2 m} = a_1^{f_2 k}$ using the above we find

$$(u_1^{e_2}u_2^{-e_1})^m = (v_1^{f_2}v_2^{-f_1})^k$$

This proves the desired equality up to signs. To see the signs work out we have to show me_1e_2 is even if and only if kf_1f_2 is even. This follows as both $me_2 = kf_2$ and $me_1 = kf_1$ (same argument as above).

Independence of choice of B. Suppose given two extensions $A \subset B$ and $A \subset B'$ as in Lemma 0EAG. Then

$$C = (B \otimes_A B')/(\mathfrak{m}\text{-power torsion})$$

will be a third one. Thus we may assume we have $A \subset B \subset C$ and factorizations over the local rings of B and we have to show that using the same factorizations over the local rings of C gives the same element of $\kappa(\mathfrak{m})$. By transitivity of norms (Fields, Lemma 0BIJ) this comes down to the following problem: if B is a Noetherian local ring of dimension 1 and $\pi \in B$ is a nonzerodivisor, then

$$\lambda^m = \prod \text{Norm}_{\kappa_k/\kappa}(\lambda)^{m_k}$$

Here we have used the following notation: (1) κ is the residue field of B, (2) λ is an element of κ, (3) $\mathfrak{m}_k \subset C$ are the maximal ideals of C, (4) $\kappa_k = \kappa(\mathfrak{m}_k)$ is the residue

field of $C_k = C_{\mathfrak{m}_k}$, (5) $m = \text{length}_B(B/\pi B)$, and (6) $m_k = \text{length}_{C_k}(C_k/\pi C_k)$. The displayed equality holds because $\text{Norm}_{\kappa_k/\kappa}(\lambda) = \lambda^{[\kappa_k:\kappa]}$ as $\lambda \in \kappa$ and because $m = \sum m_k[\kappa_k : \kappa]$. First, we have $m = \text{length}_B(B/xB) = \text{length}_B(C/\pi C)$ by Lemma 0EA9 and (0EA6). Finally, we have $\text{length}_B(C/\pi C) = \sum m_k[\kappa_k : \kappa]$ by Algebra, Lemma 02M0. □